Discrete mathematics

1. Language

Definition 1.1 (Semantics)

The meaning of a sequence of symbols.

Definition 1.2 (Syntax)

The grammatical rules for composing symbols into valid sentences.

Definition 1.3 (Variable)

A symbol that stands in place for an object that has not been determined yet.

A name can be assigned to a particular object using :=.

Definition 1.4 (Sentence)

A sentence is the expression of an idea in accordance of the syntax and grammar of a given language.

A statment is ***atomic*** if it cannot be broken down into smaller components while complying with the syntax and grammar.

- **Declarative:** Describes something. Usually, it describes a ***subject*** and a property, called ***predicate***, that the subject has.
- **Interrogative:** Asks a non-rhetorical question.
- **Imperative:** Command or request.

In math, we are concerned with declarative sentences about mathematical objects.

1.1. Church-Turing thesis

Definition 1.1.1 (Computable)

A function is ***computable*** if any of the following equivalent things are true:
- It is expressible as a general recursive process.
- It is a term in the λ-calculus.
- It can be described by a Turing machine.

Axiom 1.1.2 (Church-Turing Thesis)

Our intuitive definition of "computable" (which Turing calls "effectively calculable") is equivalent to the above definition.

2. Zeroth-order logic

2.1. Truth values

See Cedre DM1.1

Definition 2.1.1 (Truth value)

There are only two truth values:
- $\top$ "true"
- $\bot$ "false"

Definition 2.1.2 (Propositional equivalence)

Two propositions p and q are logically equivalent, denoted $p \equiv q$, iff they have the same truth value.

Definition 2.1.3

R is ***reflexive*** iff aRa for all $a \in A$.

R is ***symmetric*** iff $aRb \Leftrightarrow bRa$.

R is ***transitive*** iff $aRb, bRc \Rightarrow aRc$.

Definition 2.1.4 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Theorem 2.1.5

Logical equivalence is an equivalence relation.

Axiom 2.1.6 (Principle of Bivalence)

Sentences expressing truth values are either true or false, but not both.

Definition 2.1.7 (Proposition (informal definition))

A statement which is either true or false, i.e. a statement expressing a truth value.

2.2. Logical connectives

See Fleck 2.2~2.9

See Cedre DM1.2, CL00

Definition 2.2.1 (Negation)

$$\neg\top = \bot$$
$$\neg\bot = \top$$

Definition 2.2.2 (Logical dual)

Logical connectives f and g are logical duals iff $\neg f = g$ and $\neg g = f$ on all possible inputs.

Definition 2.2.3 (Conjunction, disjunction)

- **Conjunction:** $p \land q = p \text{ AND } q$
- **Disjunction:** $p \lor q = p \text{ OR } q$ (inclusive or)

Conjunction and disjunction are dual.

Definition 2.2.4 (Conditional)

$p \to q$ means "if p, then q". p is the ***antecedent*** and q is the ***consequent***.

If p and q are propositional formulae: $p \to q$ t is true iff every time p is satisfied, then q is satisfied.

If p and q are propositions: it is false iff p is true and q is false.

Warning: If p is always false, then $p \to q$ is always true regardless if q is false! Also, there need not be any causal relationship between p and q.

Definition 2.2.5 (Converse)

The ***converse*** of $p \to q$ is $q \to p$. These are not equivalent.

Definition 2.2.6 (Contrapositive)

The ***contrapositive*** of $p \to q$ is $\neg q \to \neg p$. It is equivalent to the original statement:

$$p \to q \equiv \neg q \to \neg p$$

Definition 2.2.7 (Biconditional)

$$p \longleftrightarrow q = p \text{ iff } q$$
$$p \longleftrightarrow q \Longrightarrow (p \to q) \land (q \to p)$$

Definition 2.2.8 (Exclusive disjunction)

$$p \oplus q = p \text{ XOR } q = p \not\leftrightarrow q$$

p	q	$\neg p$	$p \land q$	$p \lor q$	$p \oplus q$	$p \to q$	$p \longleftrightarrow q$
$\top$	$\top$	$\bot$	$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\bot$	$\top$	$\top$	$\top$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\top$

Definition 2.2.9 (Proposition)

λ is a proposition iff λ satisfies any of the following conditions:

- $\lambda = \top$ or $\lambda = \bot$.
- $\lambda = \neg\varphi$, where φ is a proposition.
- $\lambda = \varphi \land \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \lor \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \to \psi$, where φ and ψ are propositions.
- $\lambda = \varphi \longleftrightarrow \psi$, where φ and ψ are propositions.

We use this definition inductively, by first establishing that $\top$ and $\bot$ are propositions, and then using the connectives to compose larger and larger propositions. Alternatively, we use this definition recursively to determine that statements are propositions by recursively decomposing them.

2.3. Propositional formulae

See Cedre DM1.2

Definition 2.3.1 (Propositional formula)

An expression that evaluates as a proposition when all of its variables are themselves replaced by propositions.

Definition 2.3.2 (Logical equivalence and nonequivalence)

Let φ and ψ be propositional formulae both consisting of the same variables $p_1, ...p_n$. $\varphi \equiv \psi$ iff every assignment of truth values to those variables produces the same truth value.

Theorem 2.3.3

Computing a truth table to check if two propositional formulae of n variables are equivalent requires 2^n rows.

Thus, we want to prove that the two expressions are equivalent, rather than brute-forcing them.

2.4. Propositional logic

See Cedre DM1.3 *See* Fleck 2.2~2.9

Axiom 2.4.1 (Axioms of classical logic)

The following are the axioms of ***Boolean algebra***, and they have two equivalent forms:

Identity	$\top \land p \equiv p$	$\bot \lor p \equiv p$
Complement	$\neg p \land p \equiv \bot$	$\neg p \lor p \equiv \top$
Commutativity	$p \land q \equiv q \land p$	$p \lor q \equiv q \lor p$
Associativity	$p \land (q \land r) \equiv (p \land q) \land r$	$p \lor (q \lor r) \equiv (p \lor q) \lor r$
Distributivity	$p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$	

We also have the following two axioms:

- **Conditional disintegration:** $p \to q \equiv \neg p \lor q$
- **Biconditional disintegration:** $p \longleftrightarrow \equiv (p \to q) \land (q \to p)$

Theorem 2.4.2 (Uniqueness of complements)

For any p and q, if $p \land q \equiv \bot$ and $p \lor q \equiv \top$, then $\neg p \equiv q$.

Corollary 2.4.2.1

- $\top \equiv \neg\bot$ and $\bot \equiv \neg\top$
- $p \equiv q \iff \neg p \equiv \neg q$
- For any p, q, r, s s.t. $p \equiv q$ and $r \equiv s$:
  - $p \land r \equiv q \land s$
  - $p \lor r \equiv q \lor s$
  - $p \to r \equiv q \to s$
  - $p \longleftrightarrow r \equiv q \longleftrightarrow s$

Theorem 2.4.3 (Idempotence)

For any proposition p, $p \land p \equiv p$ and $p \lor p \equiv p$.

Theorem 2.4.4 (Domination)

For any proposition p, $\top \lor p \equiv \top$ and $\bot \land p \equiv \bot$.

Theorem 2.4.5 (De Morgan's Laws)

- $\neg(p \land q) \equiv \neg p \lor \neg q$
- $\neg(p \lor q) \equiv \neg p \land \neg q$

Theorem 2.4.6 (Negation equivalences)

- $\neg(\neg p) \equiv p$

- $\neg(p \to q) \equiv p \land \neg q$

2.5. Rules of inference

Definition 2.5.1 (Turnstile symbol)

$\vdash \varphi$, where φ is a predicate, means that we can prove φ with what we currently know.

$\Gamma \vdash \varphi$ means that given that assumptions Γ are true, we can prove that φ is true.

Axiom 2.5.2 (Deduction rule)

If by assuming p, we can prove q, then we can write $p \to q$:

$$(p \vdash q) \vdash (p \to q).$$

Axiom 2.5.3 (Modus ponens)

If we have $p \to q$ and we know p is true, then we can deduce q is true:

$$p, (p \to q) \vdash q.$$

Axiom 2.5.4 (Reductio ab absurdum)

If $\neg p$ leads to a contradiction, then $\neg p$ is absurd; thus p is true:

$$(\neg p \vdash q), (\neg p \vdash \neg q) \vdash p.$$

Theorem 2.5.5 (Modus tollens)

If we have $p \to q$ but also $\neg q$, then we can infer $\neg p$:

$$\neg q, (p \to q) \vdash \neg p.$$

2.5.1. Syllogisms and other theorems

Hypothetical syllogism:

$$(p \to q), (q \to r) \vdash p \to r$$

Implication elimination: (consolidation rule)

$$(p \to q) \vdash (p \vdash q)$$

Conjunction introduction: (adjunction)

$$p, q \vdash p \land q$$

Conjunction elimination: (simplification)

$$p \land q \vdash p$$

Disjunction introduction: (proof by cases)

$$p \vdash p \lor q$$

Disjunction elimination: (explosion)

$$(p \to r), (q \to r), (p \lor q) \vdash r$$

Constructive dilemma:

$$(\alpha \to \gamma), (\beta \to \delta), (\alpha \lor \beta) \vdash \gamma \lor \delta$$

3. First-order logic

See Cedre DM2, Cedre CL00, Cedre CL01

3.1. Objects

Definition 3.1.1 (Universe of discourse)

The collection of all objects that we “care about” / we “discuss”.

Definition 3.1.2 (Term)

A symbol that refers to an object in the universe of discourse.

A term which refers to a specific object is a ***constant***.

Definition 3.1.3 (Variable)

A ***variable*** is a term that refers to a generic object, not one particular object.

In order for a variable to be meaningful, it must be quantified (for example by a quantifier like $\forall$ or $\exists$) within some scope, i.e. a ***bound variable***. A variable which is not quantified is a ***free variable***.

3.2. Predicates

Definition 3.2.1 (Predicate)

Let $x_1, ..., x_n$ be variables. $\varphi(x_1, ..., x_n)$ is an ***n-ary predicate*** iff replacing each of the n variables by constants $t_1, ..., t_n$ results in a proposition $\varphi(t_1, ..., t_n)$ carrying a truth value.

3.3. Quantification

Definition 3.3.1 (Universal quantification)

The ***universal quantification*** of x appearing in φ is denoted

$$\forall x(\varphi(x, y_1, ..., y_n)).$$

It means that any constant replacing x will satisfy φ.

Definition 3.3.2 (Existential quantification)

The ***existential quantification*** of x appearing in φ is denoted

$$\exists x(\varphi(x, y_1, ..., y_n)).$$

It means that there exists some value of x which satisfies φ.

Definition 3.3.3 (Unique existential quantification)

There exists a unique x s.t. $\varphi(x)$.

$$\exists! x(\varphi(x)) \coloneqq \exists x(\varphi(x) \land y(\varphi(y) \to (y = x))).$$

3.4. Formulae

Definition 3.4.1 (Atomic formula)

A formula φ is ***atomic*** iff it satisfies one of the following:
- $\varphi = \top$ or $\varphi = \bot$.
- $\varphi = \psi(t_1, ..., t_n)$ where ψ is an n-ary predicate and $t_1, ..., t_n$ are terms.

Definition 3.4.2 (Well-formed formula)

A formula λ is ***well-formed*** (is a ***wff***) iff it satisfies one of the following:
- λ is an atomic formula.
- $\lambda = \neg\varphi$, where φ is a wff.
- $\lambda = \varphi \land \psi$, where φ and ψ are wff.
- $\lambda = \varphi \lor \psi$, where φ and ψ are wff.
- $\lambda = \varphi \to \psi$, where φ and ψ are wff.
- $\lambda = \varphi \longleftrightarrow \psi$, where φ and ψ are wff.
- $\lambda = \forall x(\varphi)$, where φ is a wff.
- $\lambda = \exists x(\varphi)$, where φ is a wff.

A well-formed formula with no free variables is called a ***sentence in the first-order logic***.

3.5. Rules of inference

Axiom 3.5.1 (Universal introduction)

If $\varphi(t)$ for an arbitrary term t, then $\forall x(\varphi(x))$.

Let t be arbitrary. Then,

$$\varphi(t) \vdash \forall x(\varphi(x)).$$

Axiom 3.5.2 (Universal elimination)

If $\forall x(\varphi(x))$, then we can pick any t and $\varphi(t)$ is true.

Choose some t. Then,

$$\forall x(\varphi(x)) \vdash \varphi(t).$$

Axiom 3.5.3 (Existential introduction)

If we know $\varphi(t)$ for a specific term t, then $\exists x(\varphi(x))$.

Choose some t. Then,

$$\varphi(t) \vdash \exists x(\varphi(x)).$$

Axiom 3.5.4 (Existential elimination)

If we know $\exists x(\varphi(x))$, then $\varphi(t)$ for some new t that has not yet appeared.

$$\exists x(\varphi(x)) \vdash \varphi(t)$$

where t comes out of the assertion.

3.6. Useful theorems

Theorem 3.6.1 (Negation of quantifiers)

For any well-formed formula φ with one free variable,

$$\neg\forall x(\varphi(x)) \equiv \exists x(\neg\varphi(x))$$
$$\neg\exists x(\varphi(x)) \equiv \forall x(\neg\varphi(x)).$$

Theorem 3.6.2 (Quantifier shift)

For any well-formed formula φ containing two free variables,

$$\forall x\forall y(\varphi(x, y)) \equiv \forall y\forall x(\varphi(x, y))$$
$$\exists x\exists y(\varphi(x, y)) \equiv \exists y\exists x(\varphi(x, y)).$$

This means that we can commute quantifiers when they are the same; however the universal and existential quantifiers don't necessarily commute with each other:

$$\exists x\forall y(\varphi(x, y)) \vdash \forall y\exists x(\varphi(x, y))$$
$$\forall x\exists y(\varphi(x, y)) \nvdash \exists y\forall x(\varphi(x, y)).$$

Theorem 3.6.3 (Distribution of quantifiers)

For any well-formed formulae φ and ψ, each containing exactly one free variable, we can distribute quantifiers as shown below:

$$\forall x(\varphi(x) \land \psi(x)) \equiv \forall x(\varphi(x)) \land \forall x(\psi(x)).$$
$$\forall x(\varphi(x) \lor \psi(x)) \equiv \exists x(\varphi(x)) \lor \exists x(\psi(x)).$$

This means that universal quantifiers can be distributed over conjunctions, and existential quantifiers can be distributed over disjunctions. The opposite does not hold.

4. Number theory

See Fleck 4, Cedre DM5

4.1. Divisibility and parity

Definition 4.1.1 (Divisibility)

For any $a, b \in \mathbb{Z}$, ***a divides b*** iff b is a multiple of a:

$$a \mid b := (\exists k \in \mathbb{Z})(b = ka)$$

We can also think about this as $a \mid b \Leftrightarrow \frac{b}{a}$ is an integer, but this definition is usually more difficult to use in proofs, and requires introducing rational numbers.

Theorem 4.1.2 (Absolutely montonicity of divisibility)

Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then, $a \mid b$ implies $|a| \leq |b|$.

Theorem 4.1.3 (Divisibility is a partial order)

The divisibility relation on $\mathbb{N}$ has the following properties:
- **Reflexive:** For all $a \in \mathbb{N}$, $a \mid a$.
- **Antisymmetric:** For all $a, b \in \mathbb{N}$, if $a \mid b$ and $b \mid a$, then $a = b$.

- **Transitive:** For all $a, b, c \in \mathbb{N}$, if $a \mid b$ and $b \mid c$, then $a \mid c$.

Theorem 4.1.4

Let $n, a, b, x, y \in \mathbb{Z}$ s.t. $n \mid x$ and $n \mid y$. Then, $n \mid ax + by$.

Definition 4.1.5 (Parity)

Let $z \in \mathbb{Z}$. z is ***even*** iff $2 \mid z$, and is ***odd*** iff $2 \mid z - 1$. For every $z \in \mathbb{Z}$, z is even xor odd.

4.2. Primality

Definition 4.2.1 (Primality)

$p \in \mathbb{N}$ is ***prime*** iff $p > 1$ and p is minimally divisible, i.e. for all $n \in \mathbb{N}$, $n \mid p$ implies $n = 1$ or $n = p$.

Any natural number other than 1 that is not prime is ***composite***.

Theorem 4.2.2 (Fundamental theorem of arithmetic)

Let $n \in \mathbb{N}$ s.t. $n \geq 2$. Then, there exists a unique $k \in \mathbb{N}$ and unique list $(p_0, a_0), ..., (p_k, a_k) \in \mathbb{N}^2$ s.t. $p_0, ..., p_k$ are distinct prime numbers satisfying:

$$n = \prod_{i=0}^{k} p_i^{\alpha_i} = p_0^{\alpha_0} p_1^{\alpha_1} ... p_k^{\alpha_k}.$$

In short, for any natural number other than 2, there exists a unique prime factorization of that number.

Theorem 4.2.3 (Euclid's Theorem)

There are infinitely many prime numbers.

4.3. GCD and LCM

Definition 4.3.1 (GCD and LCM)

The greatest common divisor of two integers $a, b \in \mathbb{Z}$ is the unique natural number $\gcd(a, b)$ such that:

- $\gcd(a, b)$ is a common divisor of a and b:
  - $\gcd(a, b) \mid a$
  - $\gcd(a, b) \mid b$
- $\gcd(a, b)$ is the greatest common divisor:
  - For all $z \in \mathbb{Z}$, $z \mid a$ and $z \mid b$ implies $z \mid \gcd(a, b)$.

We use divisibility instead of $\leq$ as our partial order, which gives us that $\gcd(0, 0) = 0$.

The least common multiple is defined dually: $\operatorname{lcm}(a, b)$ is the unique natural number such that:

- $\operatorname{lcm}(a, b)$ is a common multiple of a and b:
  - $a \mid \operatorname{lcm}(a, b)$
  - $b \mid \operatorname{lcm}(a, b)$
- $\operatorname{lcm}(a, b)$ is the least common multiple:
  - For all $z \in \mathbb{Z}$, $a \mid z$ and $b \mid z$ implies $\operatorname{lcm}(a, b) \mid z$.

We can also find GCD and LCM over finite sets of more than 2 integers in a similar manner.

Definition 4.3.2 (Coprimality)

$x, y \in \mathbb{N}$ are ***coprime*** iff $\gcd(x, y) = 1$.

Given $\mathcal{Z} \subseteq \mathbb{Z}$ and $k \in \mathbb{N} - 2$, the numbers in $\mathcal{Z}$ are ***k-wise relatively prime*** iff $\gcd(z_0, z_1, ..., z_{k-1}) = 1$ for each choice of distinct $z_0, ... z_{k-1} \in \mathcal{Z}$

Theorem 4.3.3

Given arbitrary integers $a, b \in \mathbb{Z}$, $\gcd(a, b) = 1$ iff for all $p \in \mathbb{N}$, p being prime implies that $p \nmid a$ or $p \nmid b$.

Lemma 4.3.4 (Euclid's division lemma)

If $a, b \in \mathbb{Z}$ and $b \neq 0$, there exist unique $q, r \in \mathbb{Z}$ satisfying

$$a = qb + r \text{ and } 0 \leq r < |b|.$$

We call q the ***quotient*** and r the ***remainder***.

Algorithm 4.3.5 (Euclidean division)

Given $a, b \in \mathbb{Z}$, we compute their greatest common divisor as follows.

$$\gcd(a, b) := \begin{cases} a & \text{if } b = 0 \\ \gcd(b, a \bmod b) & \text{if } b \neq 0 \end{cases}$$

Let $a \bmod b$ denote the integer r s.t. there exists a unique $q \in \mathbb{Z}$ satisfying $a = qb + r$ and $0 \leq r < |b|$.

Theorem 4.3.6 (Bézout's identity)

For all $a, b \in \mathbb{Z}$, there exists $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Theorem 4.3.7 (Euclid's lemma)

For any $a, b \in \mathbb{Z}$ and any prime $p \in \mathbb{N}$, if $p \mid ab$, then $p \mid a$ or $p \mid b$.

4.4. Congruence mod k

Definition 4.4.1 (Congruence mod k)

For any $a, b \in \mathbb{Z}$ and $k \in \mathbb{Z}^+$, $a \equiv b \pmod{k}$ iff $k \mid (a - b)$.

Definition 4.4.2 (Integers mod k)

The set of all equivalence classes under $\equiv \pmod{k}$:

$$[0]_{\equiv \pmod{k}}, [1]_{\equiv \pmod{k}}, ..., [k-1]_{\equiv \pmod{k}}$$

5. Sets

5.1. Zermelo set theory

See Cedre DM3.2

Definition 5.1.1 (First-order theory)

A formal language built upon the first-order logic by establishing a universe of discourse, introducing predicate symbols, functional symbols, and/or constant symbols, and then writing down axioms which describe the basic truths.

Functional symbols transform terms into other terms.

Definition 5.1.2 (Equality)

$$X = Y :\Leftrightarrow X \text{ and } Y \text{ refer to the same object}$$

$=$ is an equivalence relation, so it is reflexive, symmetric, and transitive.

Definition 5.1.3 (Set roster notation)

Given finitely many terms $x_0, x_1, ..., x_{n-1}$, we use $\{x_0, x_1, ..., x_{n-1}\}$ to denote the set containing exacty the objects denoted by the given terms.

For any z,

$$z \in \{x_0, x_1, ..., x_{n-1}\} :\Leftrightarrow (z = x_0) \lor (z = x_1) \lor \cdots \lor (z = x_{n-1}).$$

Definition 5.1.4 (Set builder notation)

We define $\{x : \varphi(x)\}$ to denote the set of all objects satisfying the predicate φ.

$$z \in \{x : \varphi(x)\} :\Leftrightarrow \varphi(z).$$

Definition 5.1.5 (Subset)

Given two sets A and B, A is a ***subset*** of B, denoted $A \subseteq B$, when every element of A is also an element of B.

$$A \subseteq B :\Leftrightarrow \forall z(z \in A \Rightarrow z \in B).$$

Definition 5.1.6 (Inductive set)

A set $\mathcal{I}$ is ***inductive*** if $0 \in \mathcal{I}$ and for all x, $x \in \mathcal{I}$ implies $S(x) \in \mathcal{I}$, where S is the successor function.

Axiom 0 (Infinity)

There exists an inductive set A such that for all B, if B is inductive then $A \subseteq B$.

$$\exists A(A \text{ is inductive} \land \forall B(B \text{ is inductive} \Rightarrow A \subseteq B)).$$

This set A is the “smallest” inductive set, and is equivalent to the set of natural numbers $\mathbb{N}$.

Equivalently: $\mathbb{N}$ exists.

Axiom 1 (Extensionality)

$$\forall A \forall B((A = B) \Leftrightarrow \forall z(z \in A \Leftrightarrow z \in B))$$

For all sets A and B, $A = B$ iff for all objects z, $z \in A \Leftrightarrow z \in B$.

Thus, sets are equal iff they have the same elements.

Lemma 5.1.7

For all sets A and B, $A = B$ iff $A \subseteq B$ and $B \subseteq A$.

Definition 5.1.8 (Empty set)

A set A is ***empty*** iff for all x, $x \notin A$.

We define the empty set, denoted $\emptyset$, as

$$\emptyset := \{z : z \neq z\}.$$

The empty set is empty, and it is unique.

Lemma 5.1.9

All sets contain the empty set.

Theorem 5.1.10 (Set inclusion is a partial order)

Set inclusion is a partial order, i.e. it satisfies:

- **Reflexive:** For all sets A, $A \subseteq A$.
- **Antisymmetric:** For all sets A and B, if $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- **Transitive:** For all sets A, B, and z, if $A \subseteq B$ and $B \subseteq z$, then $A \subseteq z$.

Axiom 2 (Pairing)

$$\forall A \forall B \exists z(z = \{A, B\}).$$

For all sets A and B, the set $\{A, B\}$ exists.

Russell’s paradox. Consider the set

$$A := \{x : p(x)\} \quad p(x) := x \notin x.$$

Is $A \in A$? If $A \in A$, then $p(A)$, therefore $A \notin A$. If $A \notin A$, then $\neg p(A)$, therefore $A \in A$.

Axiom 3 (Schema of Separation)

For any predicate φ with at most one free variable,

$$\forall A \exists B(B = \{z : z \in x \land \varphi(z)\}).$$

For every set A, its subset consisting of the elements of A satisfying φ exists.

Definition 5.1.11 (Power set)

Given a set A, its powerset is the set of all possible subsets of x:

$$\mathcal{P}(A) := \{B : B \subseteq A\}.$$

Axiom 4 (Power)

$$\forall A \exists \mathcal{P}(A)$$

All sets have powersets.

Lemma 5.1.12

If A is a set, $\mathcal{P}(A)$ includes $\emptyset$ and A.

Definition 5.1.13 (Union of two sets)

Given two sets A and B, the ***union*** is defined as

$$A \cup B := \{x : x \in A \vee x \in B\}.$$

This is the set which contains all the elements of A and all the elements of B, but no other elements.

Definition 5.1.14 (Union over a set)

Given a set A, the ***union over A***, i.e. the iterated union over the elements of A, is

$$\bigcup A = \bigcup_{B \in A} B := \{z : (\exists C \in A)(z \in C)\}$$

Axiom 5 (Union)

For all sets A, $\cup A$ exists.

Definition 5.1.15 (Intersection of two sets)

Given two sets A and B, the ***union*** is defined as

$$A \cup B := \{x : x \in A \wedge x \in B\}.$$

This set contains exactly the items shared by A and B.

Definition 5.1.16 (Intersection over a set)

Given a set A, the ***intersection over A***, i.e. the iterated intersection over the elements of A, is

$$\bigcap A = \bigcap_{B \in A} B := \{z : \forall C(C \in A \Rightarrow z \in C)\}$$

Definition 5.1.17 (Difference between two sets)

Given two sets A and B, the ***difference*** between A and B is defined as

$$A - B = A \setminus B := \{x : x \in A \wedge x \notin B\}.$$

Theorem 5.1.18

- For all sets A and B, $A \cup B$ exists.
- For all sets A and B, $A \cap B$ exists.
- For all sets A and B, $A - B$ exists.

Axiom 6 (Regularity)

$$\forall A(A \neq \emptyset \Rightarrow \exists B(B \in A \wedge B \cap A = \emptyset)).$$

For all nonempty sets A, there exists $B \in A$ s.t. $B \cap A$ is empty.

Theorem 5.1.19

$$\forall A(A \notin A).$$

All sets do not include themselves.

Theorem 5.1.20 (The universe does not exist)

There does not exist a "set of all sets", i.e.

- There does not exist a set U s.t. for all sets A, $A \in U$.
- $U := \{A : A \text{ is a set}\}$ does not exist.

Definition 5.1.21

$$\begin{aligned}(\forall x \in X)(\varphi(x)) &:\Leftrightarrow \forall x(x \in X \Rightarrow \varphi(x)) \\ (\exists x \in X)(\varphi(x)) &:\Leftrightarrow \exists x(x \in X \wedge \varphi(x)) \\ z \in \{x \in X : \varphi(x)\} &:\Leftrightarrow z \in X \wedge \varphi(z) \\ \{x \in X : \varphi(x)\} &:= \{x : x \in X \wedge \varphi(x)\}\end{aligned}$$

5.2. Cartesian product

Definition 5.2.1 (Ordered pair)

The ***ordered pair*** whose first coordinate is x and second coordinate is y is defined as

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

This is, more specifically, the ***Kuratowski definition of the ordered pair***.

Definition 5.2.2 (k-tuple)

The ***k-tuple*** is defined as

$$(x_1, x_2, x_3, ..., x_{k-1}) := (x_1, (x_2, (x_3, \cdots (x_{k-1}, \emptyset)))).$$

Definition 5.2.3 (Cartesian product)

The ***Cartesian product*** of two sets A and B is the set of all possible ordered pairs between them:

$$A \times B := \{(a, b) : a \in A \wedge b \in B\}.$$

5.3. Random useful theorems

Theorem 5.3.1 (DeMorgan's Laws)

$$X - (A \cup B) = (X - A) \cap (X - B)$$
$$X - (A \cap B) = (X - A) \cup (X - B)$$

Theorem 5.3.2 (Product rule)

$$|A \times B| = |A||B|.$$

6. Relations and functions

6.1. Relations

See Fleck 6

Definition 6.1.1 (Relation)

A ***relation*** $R : A \to B$ is any subset of $A \times B$.

aRb means $(a, b) \in R$.

Definition 6.1.2

R is ***reflexive*** iff aRa for all $a \in A$.

R is ***irreflexive*** iff $\neg(aRa)$ for all $a \in A$.

R is ***symmetric*** iff $aRb \Leftrightarrow bRa$.

R is ***antisymmetric*** iff $aRb \wedge bRa \Rightarrow a = b$.

R is ***transitive*** iff $aRb, bRc \Rightarrow aRc$.

Definition 6.1.3 (Equivalence relation)

A relation that is reflexive, symmetric, and transitive.

Definition 6.1.4 (Equivalence class)

Let $R : A \to B$ be an equivalence relation. Then the ***equivalence class*** of $a \in A$ is

$$[a] = [a]_R := \{b \in A : aRb\}$$

The set of all equivalence relations of R in A is called A/R.

Definition 6.1.5 (Order)

A ***partial order*** is a relation that is reflexive, antisymmetric, and transitive.

A ***linear order*** or ***total order*** is a is a partial order R in which every pair of elements are comparable.

A ***strict partial order*** is a relation that is irreflexive, antisymmetric, and transitive.

Theorem 6.1.6 (Relations can compose)

If $R : A \to B$ and $S : B \to C$, then $S \circ R : A \to C$ is a subset of $A \times C$.

⚠ Warning

$R \circ R^{-1}$ is not necessarily the identity map!

Theorem 6.1.7

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}$$
$$(T \circ S) \circ R = T \circ (S \circ R)$$

6.2. Functions

Definition 6.2.1 (Function)

A ***function*** (also called a ***map*** or ***mapping***) is a relation $f : A \to B$ where each $a \in A$ appears as the first element (input) of an ordered pair $(a, b) = afb$ exactly once.

If $(a, b) \in f$, we can also say $f(a) = b$. b is the ***value*** of f at a, or the ***image*** of a under f.

A is the ***domain*** and B the ***codomain*** of f.

Definition 6.2.2 (Injective)

A function f is ***injective*** or ***one-to-one*** iff:
- $f(a) = f(b) \Rightarrow a = b$, or equivalently
- $a \neq b \Rightarrow f(a) \neq f(b)$

Definition 6.2.3 (Surjective)

A function $f : A \to B$ is ***surjective*** or ***onto*** iff:
- $f(A) = B$ (its image is its codomain), or equivalently
- for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$

Definition 6.2.4 (Bijective)

A function is ***bijective*** and has ***one-to-one correspondence*** if it is both injective and surjective.

Theorem 6.2.5

A composition of two functions is a function. If both are sur/in/bijective, so is their composition.

Theorem 6.2.6

Strictly increasing functions in $\mathbb{R} \to \mathbb{R}$ are bijective.

7. Induction

7.1. Least element principle

Definition 7.1.1 (Order on $\mathbb{N}$)

$$x < y :\Leftrightarrow x \in y$$
$$x \leq y :\Leftrightarrow (x < y) \lor (x = y)$$

$\leq$ forms a total order on $\mathbb{N}$, and $<$ forms a strict total order on $\mathbb{N}$.

Theorem 7.1.2 (Least element property of $\mathbb{N}$)

For all nonempty subsets M of the naturals, there exists $m \in M$ s.t. for all $n \in M$, $m \leq n$.

m is then the minimum of M.

Theorem 7.1.3 (Weak induction)

For any wff φ with at most one free variable, the following holds:

$$(\forall n \in \mathbb{N})(\varphi(n)) \Leftrightarrow \varphi(0) \land (\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(\operatorname{succ}(k))).$$

Theorem 7.1.4 (Strong induction)

For any wff φ with at most one free variable, the following holds:

$$(\forall n \in \mathbb{N})(\varphi(n)) \Leftrightarrow \varphi(0) \land (\forall k \in \mathbb{N})$$
$$((\varphi(0) \land \varphi(1) \land \dots \land \varphi(k)) \Rightarrow \varphi(\operatorname{succ}(k))).$$

Any proof using strong induction has an equivalent proof using weak induction.

7.2. Primitive arithmetic operations on the naturals

Definition 7.2.1 (Addition)

$$n + 0 := n$$
$$n + \operatorname{succ}(m) := \operatorname{succ}(n + m) \text{ if } m \in \mathbb{N}$$

Definition 7.2.2 (Multiplication)

$$\begin{aligned} n * 0 &:= n \\ n * \mathrm{succ}(n) &:= (n * m) + n \text{ if } m \in \mathbb{N} \end{aligned}$$

Definition 7.2.3 (Exponentiation)

$$\begin{aligned} n^0 &:= 1 \\ n^{\mathrm{succ}(m)} &:= n^m * n \text{ if } m \in \mathbb{N} \end{aligned}$$

Definition 7.2.4 (Exponentiation)

$$\begin{aligned} n \uparrow\uparrow 0 &:= 1 \\ n \uparrow\uparrow \mathrm{succ}(m) &:= n^{n \uparrow\uparrow m} \text{ if } m \in \mathbb{N} \end{aligned}$$

Definition 7.2.5 (Iterated addition)

Let $f : \mathbb{N} \to \mathbb{N}$ be a sequence of natural numbers. Then,

$$\begin{aligned} \sum_{i=a}^{b} f(i) &:= 0 && \text{if } b < a \\ \sum_{i=a}^{a} f(i) &:= f(a) && \text{if } b = a \\ \sum_{i=a}^{\mathrm{succ}(b)} f(i) &:= \left(\sum_{i=a}^{b} f(i) \right) + f(\mathrm{succ}(b)) && \text{if } b > a \end{aligned}$$

Definition 7.2.6 (Iterated addition)

Let $f : \mathbb{N} \to \mathbb{N}$ be a sequence of natural numbers. Then,

$$\begin{aligned} \sum_{i=a}^{b} f(i) &:= 0 && \text{if } b < a \\ \sum_{i=a}^{a} f(i) &:= f(a) && \text{if } b = a \\ \sum_{i=a}^{\mathrm{succ}(b)} f(i) &:= \left(\sum_{i=a}^{b} f(i) \right) + f(\mathrm{succ}(b)) && \text{if } b > a \end{aligned}$$

Definition 7.2.7 (Iterated multiplication)

Let $f : \mathbb{N} \to \mathbb{N}$ be a sequence of natural numbers. Then,

$$\begin{aligned} \prod_{i=a}^{b} f(i) &:= 0 && \text{if } b < a \\ \prod_{i=a}^{a} f(i) &:= f(a) && \text{if } b = a \\ \prod_{i=a}^{\mathrm{succ}(b)} f(i) &:= \left(\prod_{i=a}^{b} f(i) \right) * f(\mathrm{succ}(b)) && \text{if } b > a \end{aligned}$$

7.3. Cardinality

Definition 7.3.1 (Relative cardinality)

$|X| \leq |Y|$ iff there exists an injection $X \to Y$.

$|X| \geq |Y|$ iff $Y = \emptyset$ or there exists a surjection $X \to Y$.

$|X| = |Y|$ iff there exists a bijective function $X \to Y$.

Definition 7.3.2 (Left-inverse)

$f : X \to Y$ is ***left-invertible*** iff there exists $g : Y \to X$ s.t. $g \circ f = \mathrm{id}_X$.

Left-invertible implies monomorphism.

Definition 7.3.3 (Monomorphism)

$f : X \to Y$ is a ***monomorphism***, denoted $f : X \hookrightarrow Y$, iff for any set Z and any pair of functions $g_1 : X \to Z$ and $g_2 : X \to Z$, we have $f \circ g_1 = f \circ g_2 \Rightarrow g_1 = g_2$.

Definition 7.3.4 (Right-inverse)

$f : X \to Y$ is ***right-invertible*** iff there exists $g : Y \to X$ s.t. $f \circ g = \mathrm{id}_Y$.

Right-invertible implies epimorphism.

Definition 7.3.5 (Epimorphism)

$f : X \to Y$ is a ***epimorphism***, denoted $f : X \twoheadrightarrow Y$, iff for any set Z and any pair of functions $g_1 : X \to Z$ and $g_2 : X \to Z$, we have $f \circ g_1 = f \circ g_2 \Rightarrow g_1 = g_2$.

Definition 7.3.6 (Isomorphism)

$f : X \to Y$ is an ***isomorphism*** iff there exists a function $g : Y \to X$ s.t. $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. If such a g exists, it will be unique, and is denoted f^{-1}, the ***inverse*** of f.

Axiom 7 (Trichotomy of Cardinality)

For any sets A and B, we have $|A| < |B|$ and $|A| > |B|$ or $|A| = |B|$.

This is equivalent to the axiom of choice.

Theorem 7.3.7 (Cantor-Schröder-Bernstein)

For all sets X and Y, $|X| \leq |Y|$ and $|Y| \leq |X|$ iff $|X| = |Y|$.

Theorem 7.3.8

Every injection has a surjective left-inverse, and is therefore a monomorphism.

Every surjection has an injective right-inverse, and is therefore an epimorphism.

Every bijection has a unique two-sided inverse, and is therefore an isomorphism.

$\leq$, $\geq$, and $=$ behave as expected, with regards to cardinalities.

Theorem 7.3.9

$X \subseteq Y$ implies $|X| \leq |Y|$.

Theorem 7.3.10 (Pigeonhole Principle)

For any sets A and B:
- If $|A| > |B|$, there are no injective functions $A \to B$.
- If $|A| < |B|$, there are no surjective functions $A \to B$.

If A and B are both finite, then for any function $f : A \to B$, there exists $b \in B$ s.t.

$$|\{a \in A : f(a) = b\}| \geq \left\lceil \frac{|A|}{|B|} \right\rceil = \left\lfloor \frac{|A|-1}{|B|} \right\rfloor + 1.$$

7.4. Finite combinatorics

Definition 7.4.1 (Finite)

A set A is finite iff there exists $n \in \mathbb{N}$ s.t. $|F| = |n|$.

Definition 7.4.2 (Cardinality of finite set)

The ***cardinality*** of a finite set A, denoted $|A|$, is the number of elements in the set. Equivalently, it is the natural number n that A is in bijection with: $|A| = |n|$.

Lemma 7.4.3

- Every natural number is a finite set.
- $|\{z \in \mathbb{N} : 1 \leq z \leq n\}| = |n|$ for all $n \in \mathbb{N}$.
- For any finite set A and any set B, if $B \subseteq A$, then B is finite.
- If A and B are finite sets, then $|A \cup B| = |A| + |B| - |A \cap B|$.
- For any finite set X, $|\mathcal{P}(X)| = 2^{|X|}$.

Theorem 7.4.4 (Inclusion-exclusion principle)

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Definition 7.4.5 (Binomial coefficients)

For any $k, n \in \mathbb{N}$, $\binom{n}{k}$ is the number of subsets of cardinality k taken from a set of cardinality n:

$$\binom{n}{k} := |\{A : A \subseteq n \land |A| = k\}|.$$

Lemma 7.4.6 (Recursive calculation of binomial coefficients)

For all $n, k \in \mathbb{N}$:
- $\binom{n}{0} = 1 = \binom{n}{n}$
- $\binom{n}{1} = n$
- $\binom{n+1}{k+1} = \binom{n}{k+1} + \binom{n}{k}$.

Lemma 7.4.7

For all $n, k \in \mathbb{N}$ s.t. $k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}.$$

Theorem 7.4.8 (Binomial theorem)

For any real numbers $x, y \in \mathbb{R}$ and any natural $n \in \mathbb{N}$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

7.5. Infinite combinatorics

Definition 7.5.1 (Infinity)

A set I is ***infinite*** iff it is not finite.

Theorem 7.5.2

- $\mathbb{N}$ is infinite.
- $|\mathbb{N}| = |\mathbb{N}^+| = |\mathbb{Z}| = |\mathbb{Q}| = |\mathbb{N} \times \mathbb{N}|$.

Theorem 7.5.3

For any set Y and infinite set X,
- $\operatorname{card}(X \cup Y) = \max(|X|, |Y|)$.
- $\operatorname{card}(X \times Y) = \max(|X|, |Y|)$.
- If $2 \leq |Y| \leq |X|$, then $|\{f : (f : X \to Y)\}| = |\{f : (f : X \to \{0,1\})\}| = |\mathcal{P}(X)|$.

Definition 7.5.4 (Countability)

A set A is ***countable*** iff $|A| \leq |\mathbb{N}|$, otherwise it is ***uncountable***.

A set A is ***countably infinite*** iff
- it is countable and infinite, or equivalently
- $|A| = |\mathbb{N}|$

Thus, $\mathbb{N}$ is the smallest infinite set.

Theorem 7.5.5

- Every subset of $\mathbb{N}$ is countable.
- Every finite set is countable.
- The countable union of countable sets is countable.

Definition 7.5.6 (Dedekind infinity)

The set X is ***Dedekind finite*** iff for all sets $W \subsetneq X$, $|W| < |X|$. This is equivalent to being finite.

The set X is ***Dedekind infinite*** iff there exists a set $W \subsetneq X$ s.t. $|W| = |X|$. This is equivalent to being infinite.

Theorem 7.5.7 (Cantor's theorem)

For any set X, $|X| < |\mathcal{P}(X)|$.